Gauge Structure from Admissible Non-Injective Projection

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Abstract

Starting from the axiomatic foundation of admissible non-injective transitions [**?**], we derive the structure of gauge theory without introducing it as a postulate. Gauge charges are identified as invariants of equivalence classes in the projection fibre: two configurations are gauge-equivalent when they project to the same observable state. The group of fibre-preserving transformations constitutes the gauge group, with U(1), SU(2), and SU(3) identified as natural candidates from the admissibility structure. The gauge connection arises as the correction required to compare admissible fibres over neighbouring points, and Yang–Mills dynamics follow from minimising the curvature of this connection. No dynamical substrate is required at any step; the derivations rest solely on Axioms A1–A4 of [**?**] and the spectral results of the O-series. Effective spacetime geometry is derived in the companion paper [**?**].

1 Introduction

Position in the programme

The companion papers [**?**, **?**, **?**] establish the quantum-mechanical structure of the Cosmochrony framework: the proto-state, phase coherence, the Born rule, the singlet correlator, and the Tsirelson bound, all derived from the admissibility structure of non-injective projection. These results concern the *internal* structure of the admissible fibre at a single transition step.

The present paper addresses one macroscopic consequence of allowing the projection structure to vary from point to point in the effective continuum: the conserved invariants of the projection fibre and the symmetry group they generate. The answer is gauge charges and gauge groups (§**??**), with Yang–Mills dynamics emerging from fibre transport (§**??**). Effective spacetime geometry is derived in the companion paper [**?**].

What is and is not derived in this paper

This paper derives the *structure* of gauge theory from the admissibility framework. It does not derive specific coupling constants or the precise matter content of the Standard Model. These remain open and are identified as directions for future work in Section **??**.

The derivations are structural: they identify which gauge groups are *admissible* given the constraints, not which specific values their parameters take.

Organisation

Section **??** derives gauge charges as fibre invariants and identifies the gauge group. Section **??** derives the gauge connection from the requirement of consistent transport between neighbouring fibres. Section **??** derives Yang–Mills dynamics from the minimisation of connection curvature. Section **??** identifies the open directions.

2 Gauge Charges as Fibre Invariants

2.1 Charges from non-injectivity

The non-injectivity of Π_n (Axiom A2 of [**?**]) means that multiple configurations in $\Omega_n^{(c)}$ project to the same observable state O_n. These configurations are physically indistinguishable at the observable level but may carry distinct internal structure.

Definition 2.1 (Gauge charge)**.** *A* gauge charge *is a functional*

$$Q : \Omega_n^{(c)} \longrightarrow \mathbb{R} \tag{1}$$

that is constant on projection fibres:

$$\Pi_n(\omega_1) = \Pi_n(\omega_2) \implies Q(\omega_1) = Q(\omega_2). \tag{2}$$

A gauge charge is therefore an observable that cannot be resolved by Π_n itself — it lives in the kernel structure of Π_n — but is well-defined on the fibre.

Definition 2.2 (Gauge group)**.** *The* gauge group $\mathcal{G}$ *is the group of transformations $\omega \mapsto g \cdot \omega$ that preserve the projection:*

$$\mathcal{G} := \big\{ g \in \mathrm{Aut}(\Omega_n^{(c)}) \mid \Pi_n(g \cdot \omega) = \Pi_n(\omega) \text{ for all } \omega \big\}. \tag{3}$$

$\mathcal{G}$ consists precisely of the fibre-preserving symmetries: it acts within $\Pi_n^{-1}(O_n)$ without affecting the observable output.

Remark 2.3 (Gauge invariance as structural necessity)**.** *The gauge group is not introduced by hand. It is forced by the structure of* Π_n*: as soon as* Π_n *is non-injective, there exist non-trivial fibre-preserving transformations, and hence a non-trivial gauge group. Gauge invariance is a structural consequence of Axiom A2, not an additional postulate.*

2.2 Identification of the gauge groups

Three natural candidates emerge from the fibre structure established in Q1 [**?**] and Q2 [**?**]:

- U(1): global phase rotations $\omega \mapsto e^{i\theta}\omega$ within the complex fibre $\mathbb{C}^q$. These preserve $|\omega|$ and hence the observable norm, while acting non-trivially on the phase — which is in the kernel of Π_n by the phase invariance of $\sigma_c(n)$ [**?**].
- SU(2): the stable fixed point of the admissibility flow established in [**?**] Theorem 8.1. Its fibre-preserving action on $\mathbb{C}^2$ is the spin-$\frac{1}{2}$ structure derived in [**?**].
- SU(3): a structurally motivated candidate for a higher-dimensional internal symmetry of the fibre, corresponding to multiple co-admissible internal directions beyond SU(2). Its identification from the O-series admissibility data is an open direction (Section **??**).

Remark 2.4 (Status of SU(3))**.** *The identification of* SU(3) *as a gauge group is a structural hypothesis, not a derived result. Establishing it would require finding, within the O-series spectral data, a co-admissibility pattern for a group with* SU(3) *structure analogous to the* $\lambda_{1/2} = \lambda_{3/2}$ *co-admissibility of [**?**]. This is identified as a medium-term open direction.*

3 The Gauge Connection

3.1 The problem of comparing neighbouring fibres

The following development assumes that the admissible fibre F_n admits an effective continuum description in which it forms a smooth fibre bundle over an effective spacetime manifold. The precise sense in which this limit holds is established conditionally in [**?**]; it is used here as a working hypothesis.

For each point x in the effective manifold, the fibre $F_x = \Pi^{-1}(x)$ carries the internal gauge structure of Section **??**.

When the projection $\Pi(x)$ varies from point to point, the fibres F_x and F_{x+dx} are not canonically identified. A state $\psi(x)$ transforming under $\mathcal{G}$ acquires an additional term under differentiation:

$$\partial_\mu(g(x)\,\psi(x)) \;=\; g(x)\,\partial_\mu\psi(x) + (\partial_\mu g(x))\,\psi(x), \tag{4}$$

which is not covariant under local gauge transformations $g(x) \in \mathcal{G}$.

3.2 Gauge connection from fibre transport

Definition 3.1 (Gauge connection)**.** *A* gauge connection A_μ *is the correction required to define a covariant derivative*

$$D_\mu \;:=\; \partial_\mu + A_\mu \tag{5}$$

such that $D_\mu\psi$ transforms homogeneously under local gauge transformations.

Proposition 3.2 (Admissibility interpretation of A_μ)**.** *In the projection framework, $A_\mu(x)$ encodes the local variation of the admissible fibre structure: it is the infinitesimal correction needed to transport an admissible configuration from the fibre at x to the fibre at $x + dx$ consistently with the admissibility constraints.*

Structural argument. The admissibility constraints (A1–A3 of [**?**]) define, at each point x, a fibre F_x of structurally compatible configurations. When $\Pi(x)$ varies, the fibres F_x and F_{x+dx} are related by the differential of the projection map. Consistency of the admissibility condition across neighbouring fibres — i.e. that a configuration admissible at x maps to an admissible configuration at $x + dx$ — requires a linear correction of the form $A_\mu(x)\, dx^\mu$. This correction is precisely the gauge connection. □

Remark 3.3 (Relation to the holonomy argument)**.** *The white paper [**?**] establishes that non-injective admissible projections generically have non-trivial holonomy, which is the minimal structural condition for a gauge symmetry to arise as a descriptive invariance. Proposition* **??** *is the local infinitesimal version of this global statement.*

4 Yang–Mills Dynamics

4.1 Field strength as curvature of the connection

Definition 4.1 (Field strength)**.** *The* field strength *associated with the gauge connection A_μ is the curvature of the covariant derivative:*

$$F_{\mu\nu} \;:=\; [D_\mu, D_\nu] \;=\; \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu]. \tag{6}$$

$F_{\mu\nu}$ measures the failure of commutativity of parallel transport around an infinitesimal loop — equivalently, the non-commutativity of the admissibility constraints in two neighbouring directions.

4.2 Yang–Mills action from admissibility minimisation

Proposition 4.2 (Yang–Mills action from coherence minimisation)**.** *The natural dynamical principle for the gauge connection is minimisation of the deformation of the projection structure, leading to the action*

$$S_{\mathrm{YM}}[A] \;=\; \int \mathrm{Tr}\big(F_{\mu\nu}F^{\mu\nu}\big)\,\sqrt{-g}\,d^4x. \tag{7}$$

Structural argument. By Proposition **??**, A_μ encodes the local variation of the fibre structure. The curvature $F_{\mu\nu}$ measures the incompatibility of this variation around closed loops: a flat connection ($F_{\mu\nu} = 0$) corresponds to a globally consistent fibre structure, while non-zero curvature signals inconsistency. The admissibility principle (A3 of [**?**]) favours configurations that preserve the full multiplicity of open directions without premature selection. Translated to the gauge connection, this principle selects configurations that minimise incompatibility, i.e. minimise $\|F_{\mu\nu}\|^2$. The action (**??**) is the unique gauge-invariant, Lorentz-invariant, second-order functional of A_μ that achieves this. □

Corollary 4.3 (Yang–Mills equations)**.** *Variation of* (**??**) *with respect to* A_μ *yields*

$$D_\mu F^{\mu\nu} \;=\; 0, \tag{8}$$

the Yang–Mills equations in vacuum.

Remark 4.4 (Status)**.** *Proposition* **??** *is a structural motivation, not a complete proof. The uniqueness claim requires showing that* (**??**) *is the* only *admissibility-compatible action at second order in* A_μ*. This follows in the Born–Infeld limit from the results of [**?**] (the BI action reduces to the Yang–Mills action in the weak-field regime), but a direct derivation from the axioms of [**?**] alone is an open step identified in Section* **??**.

5 Open Directions

Direct derivation of Yang–Mills uniqueness. Proposition **??** gives a structural motivation for the Yang–Mills action. Its uniqueness — that S_{YM} is the *only* admissibility-compatible second-order action — follows from the Born–Infeld limit [**?**] but not yet from Foundation M axioms alone.

SU(3) **from admissibility.** The identification of SU(3) as a gauge group requires finding a co-admissibility pattern for an SU(3)-type group in the O-series spectral data. The analogue of the $\lambda_{1/2} = \lambda_{3/2}$ result on $2I$ (established in [**?**]) must be found for a group with SU(3) structure.

Coupling constants. The present paper identifies which gauge groups are admissible. It does not determine the numerical values of coupling constants $(g_s,\ g_w,\ \alpha)$. These are expected to emerge from the quantitative spectral data of the O-series via the transfer chain $c_\chi \to \delta_{\text{pair}} \to \beta^* \approx 0.126$, but the explicit identification is open.

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